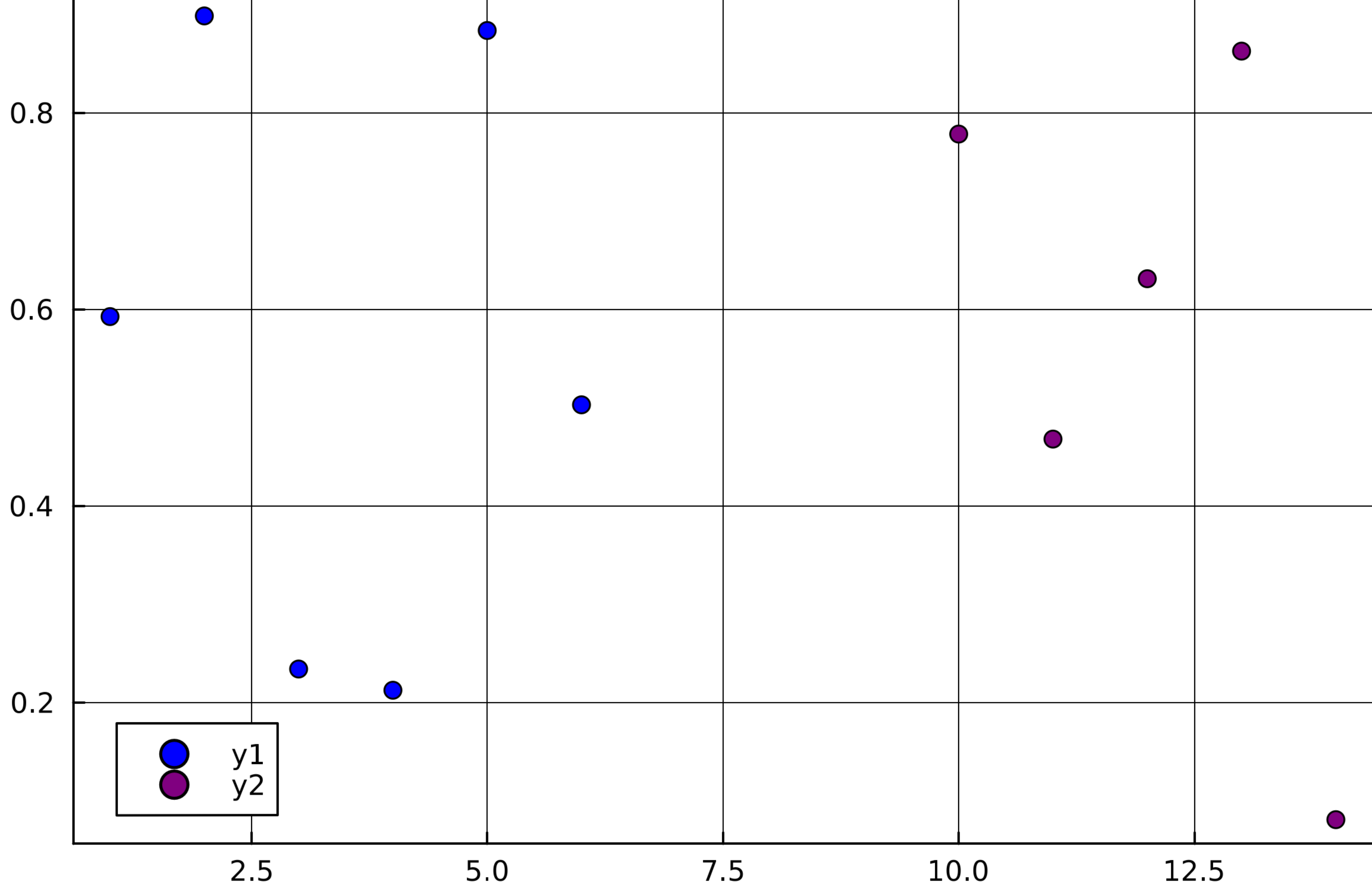


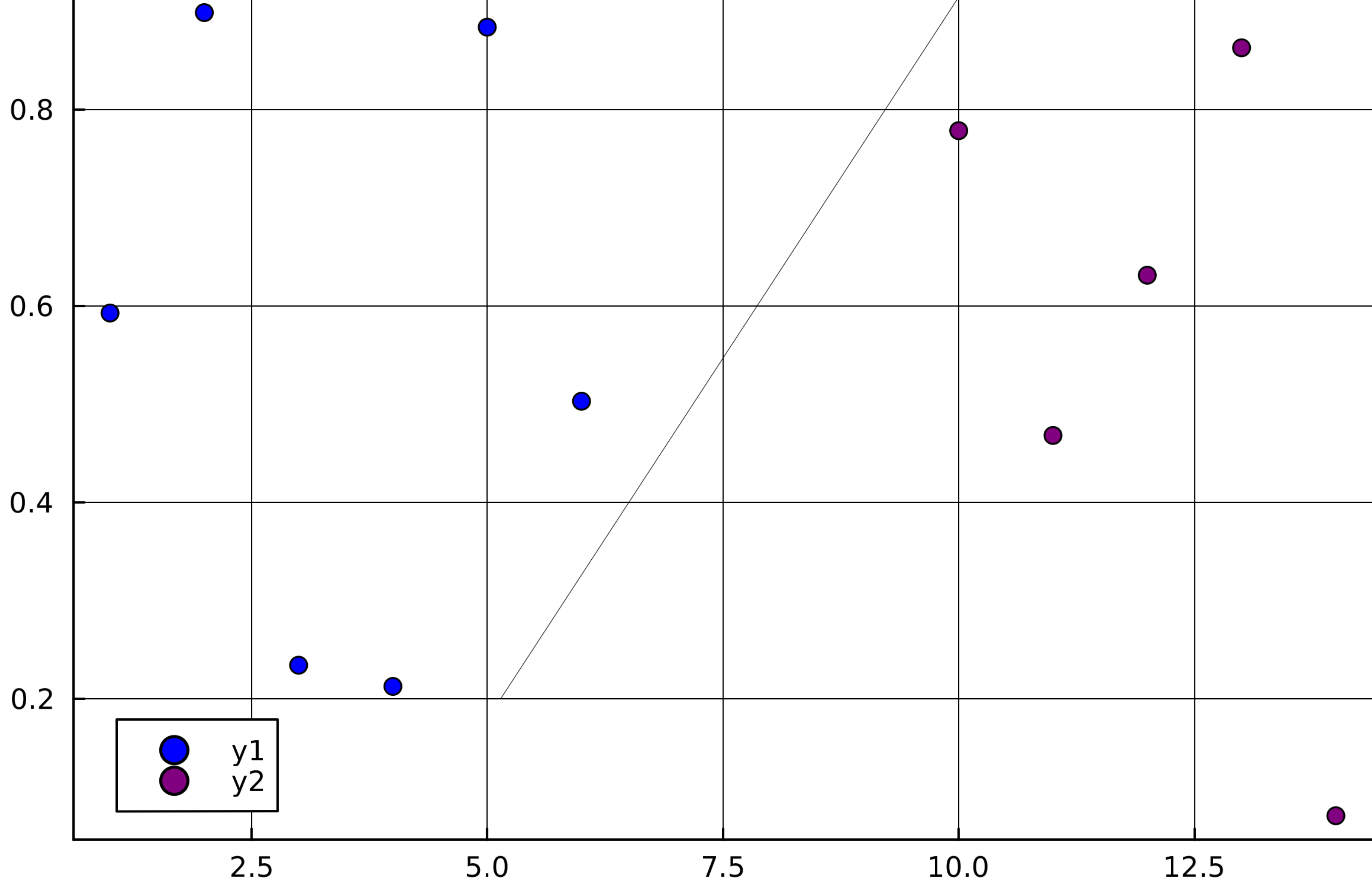
ta911s

José Quenum
PhD Computer Science



support vector machines





- illustrate with multiple decision boundaries

goal

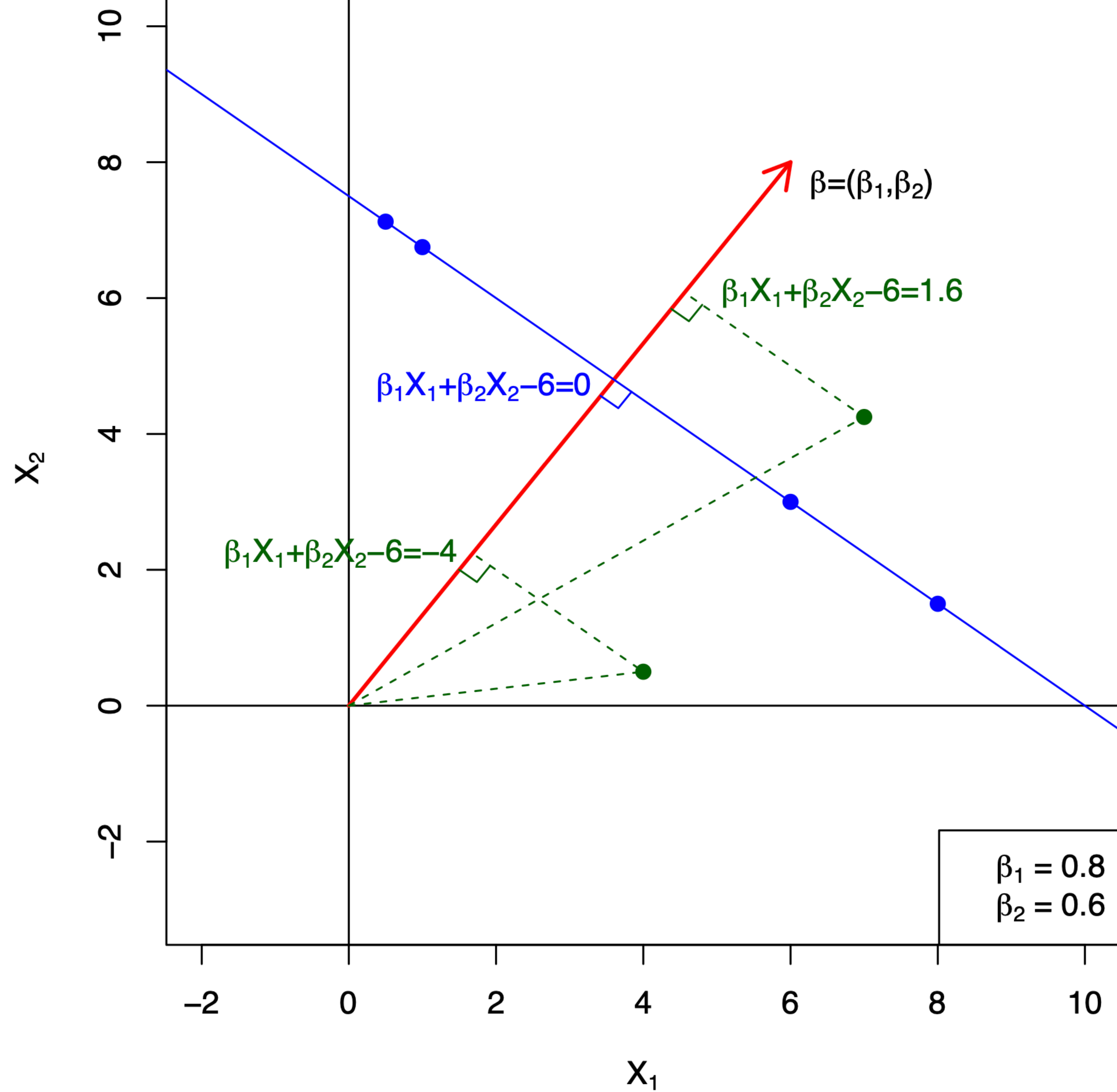
- define a **hyperplane** that separates the classes in feature space
- define a **hyperplane** that fits data points in a continuous space

classification

$$\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p = 0$$

- in $\text{dim} = 1$, hyperplane is a point
- in $\text{dim} = 2$, hyperplane is a line

$\vec{\beta} = (\beta_1, \beta_2, \dots, \beta_p)$ is a normal vector, i.e., orthogonal to the hyperplane



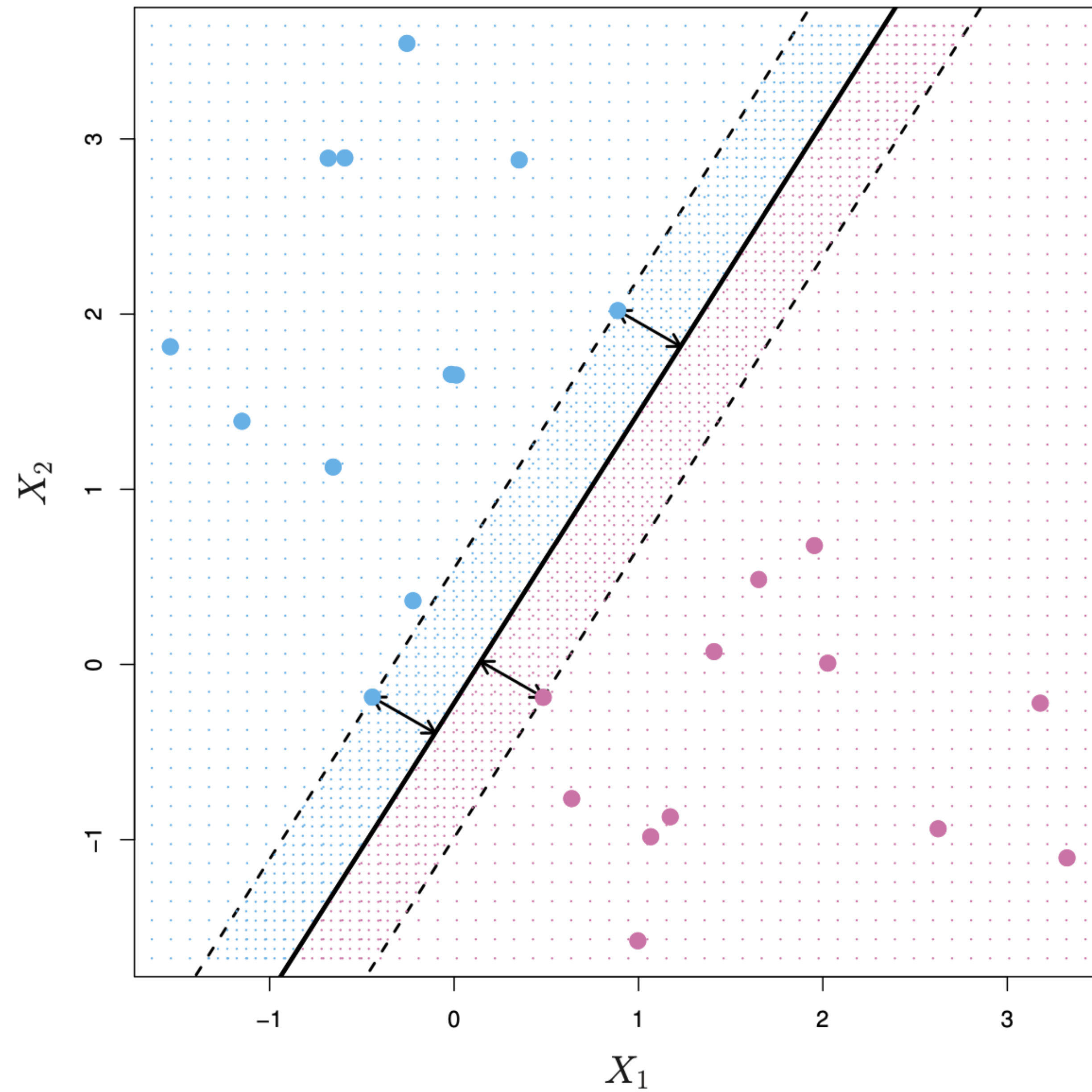
maximal margin classifier

- what is the separating hyperplane?

$$f(x) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$$

$$\begin{cases} f(x) > 0 \\ f(x) < 0 \end{cases}$$

hyperplane: $f(x) = 0$



maximize

$$\beta_0, \beta_1, \dots, \beta_p$$

subject to

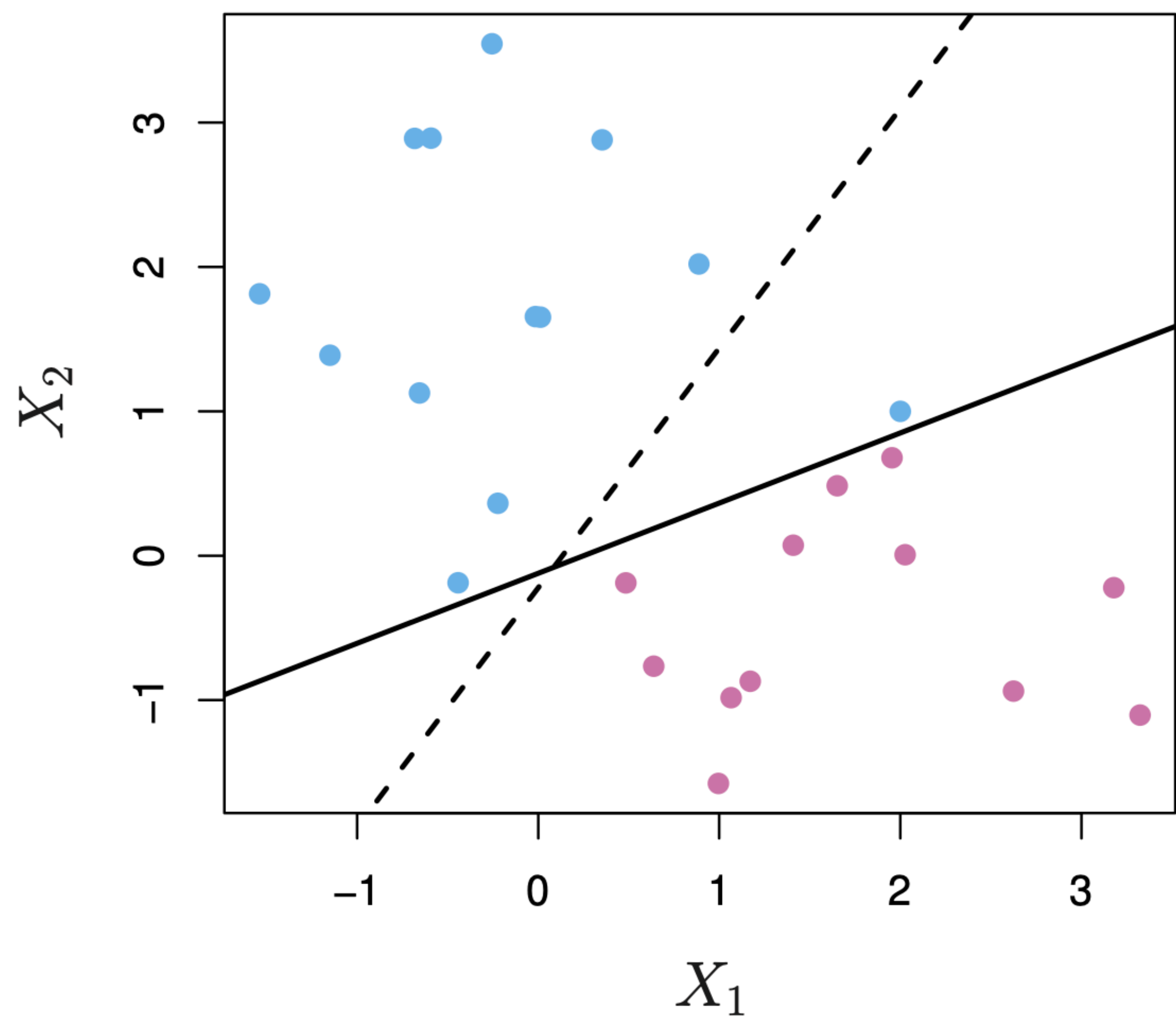
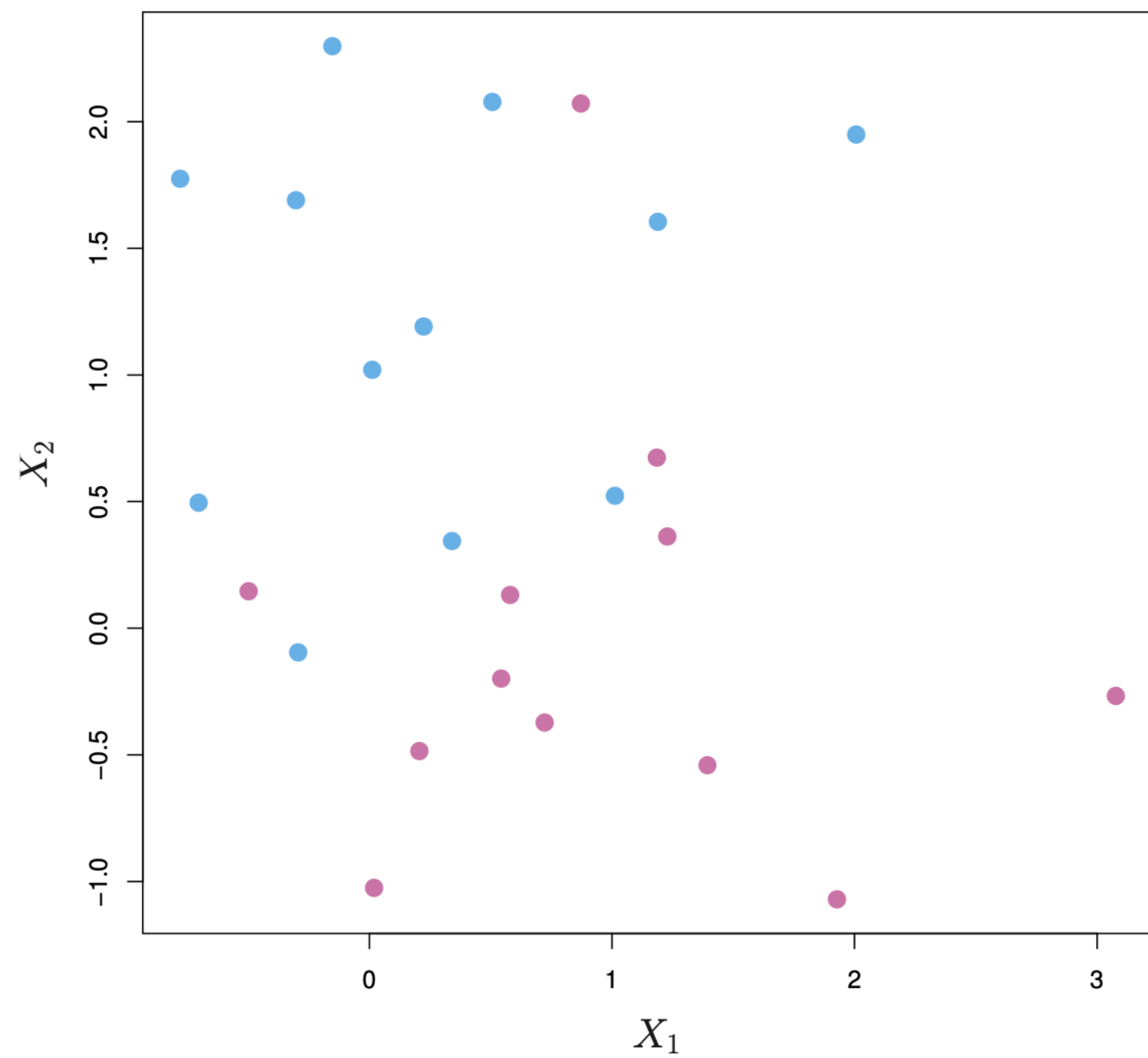
$$M$$

$$\sum_{j=1}^p \beta_j^2 = 1$$

$$y_l(\beta_0 + \beta_1 x_{l1} + \beta_2 x_{l2} + \dots + \beta_p x_{lp}) \geq M$$

$$\forall l \in 1..N.$$

- non separable data
- noisy data



support vector classifier

maximize

$$\beta_0, \beta_1, \dots, \beta_p$$

subject to

$$M$$

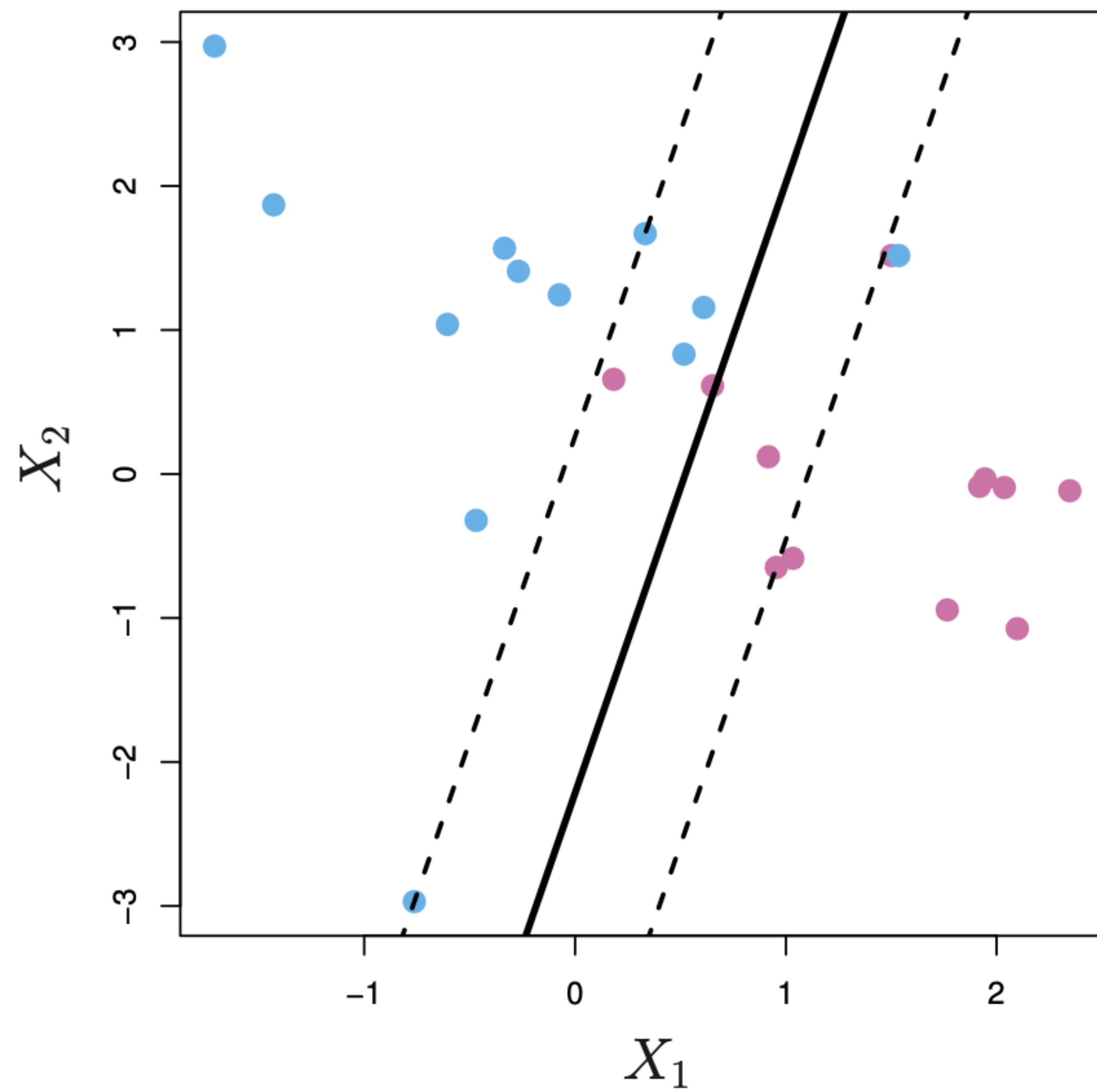
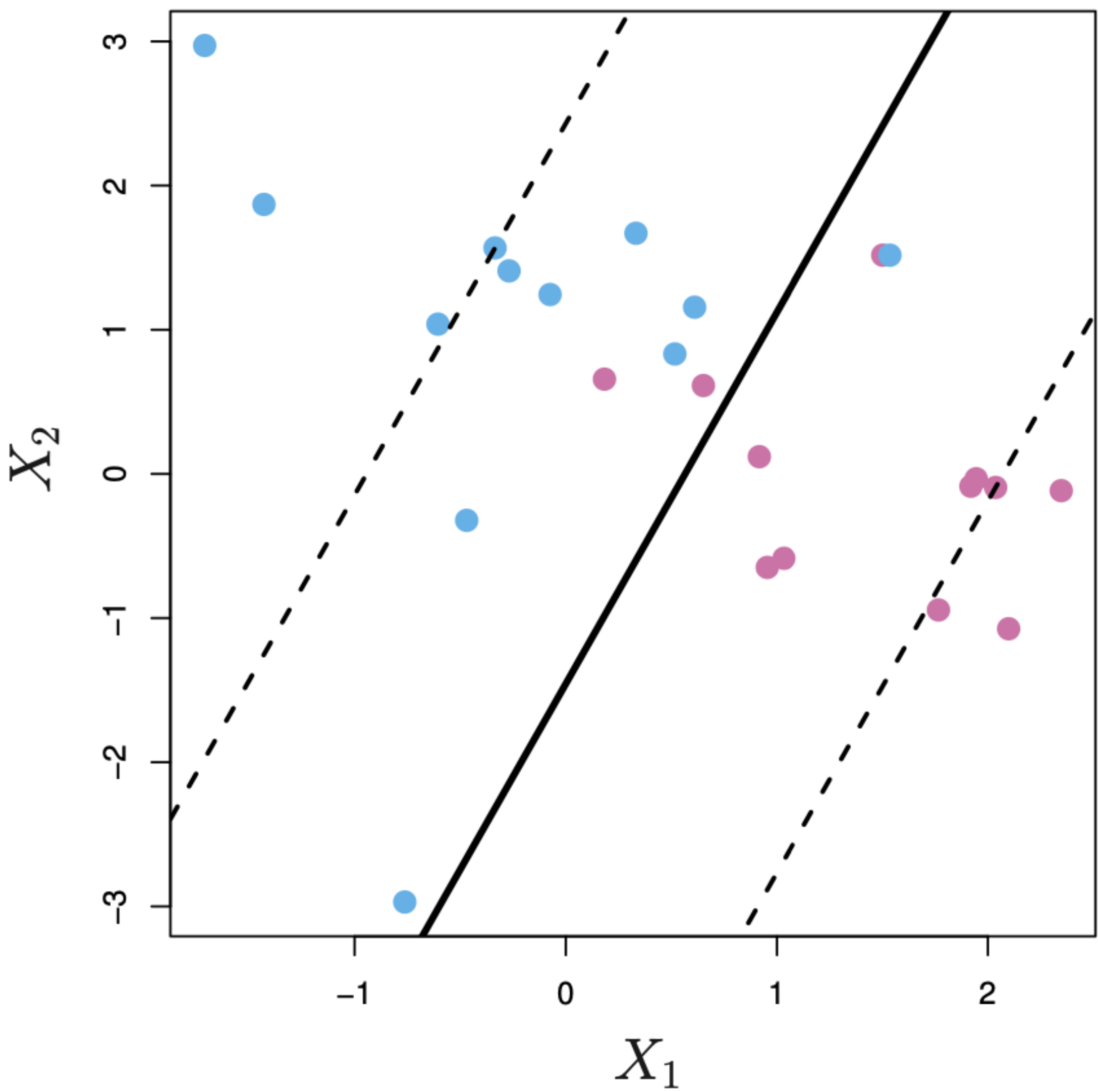
$$\sum_{j=1}^p \beta_j^2 = 1$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}) \geq M(1 - \epsilon_i)$$

$$\epsilon_i \geq 0; \quad \sum_{i=1}^N \epsilon_i \leq C$$

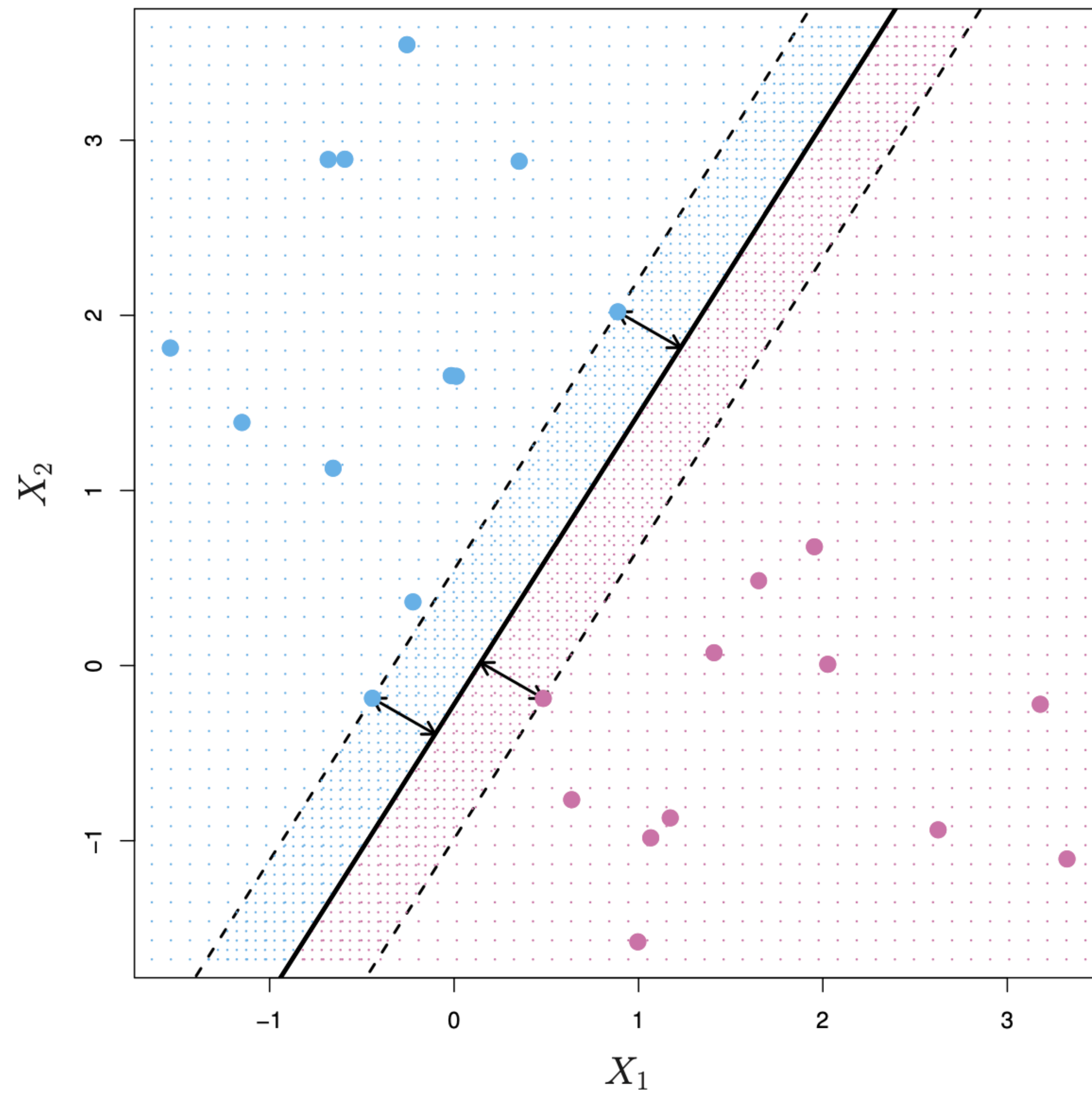
$$\begin{cases} \epsilon_l = 0 \implies \text{observation on the correct side of the margin} \\ \epsilon_l > 0 \implies \text{observation on the wrong side of the margin} \\ \epsilon_l > 1 \implies \text{observation on the wrong side of the hyperplane} \end{cases}$$

- **C determines #violations that can be tolerated**
- **small C implies narrow margin (less violations)**
- **larger C implies wider margin (more violations)**

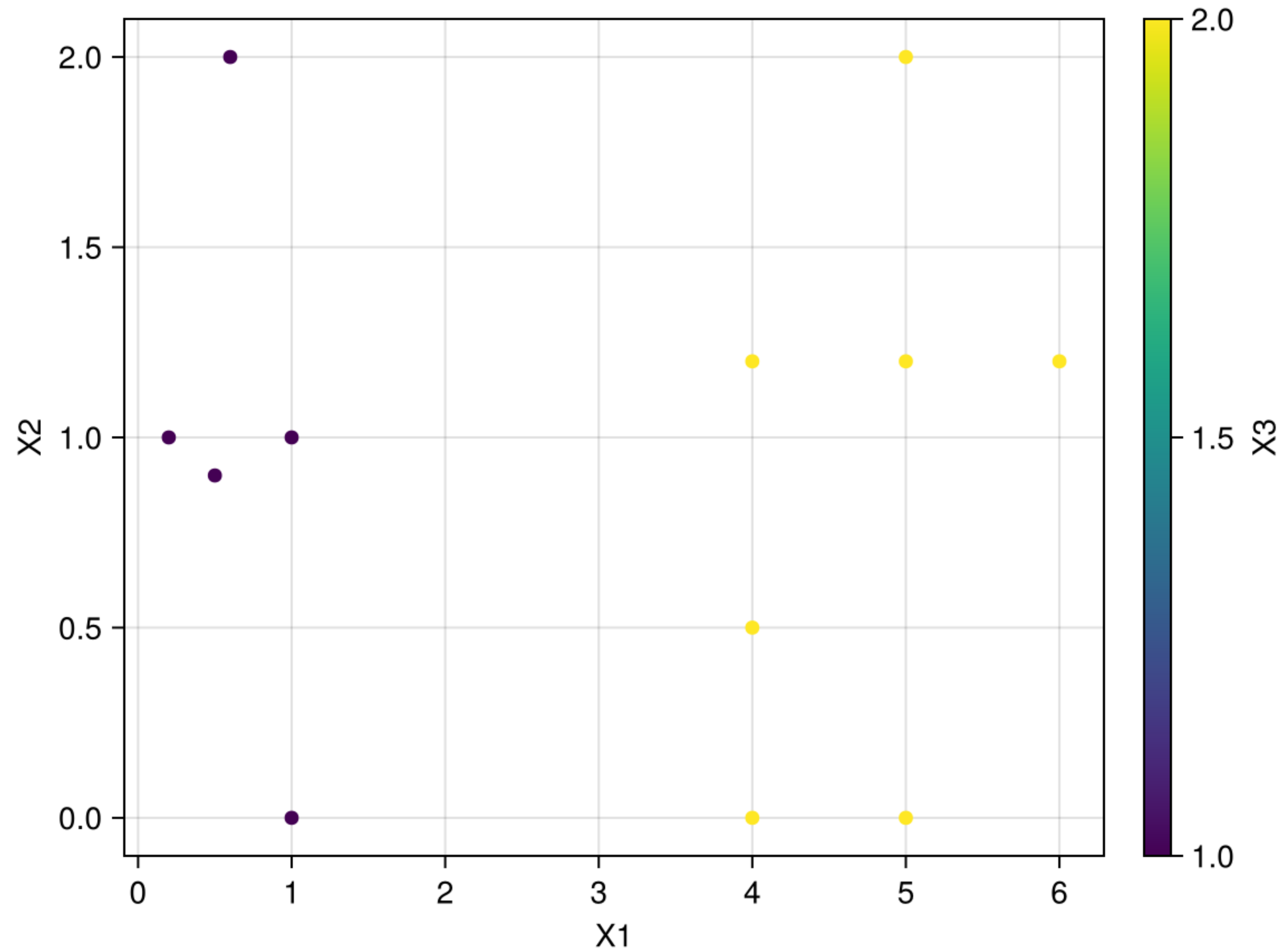


important!!!

- observations that lie on the margin or violate the margin affect the hyperplane
- those observations are the support vector

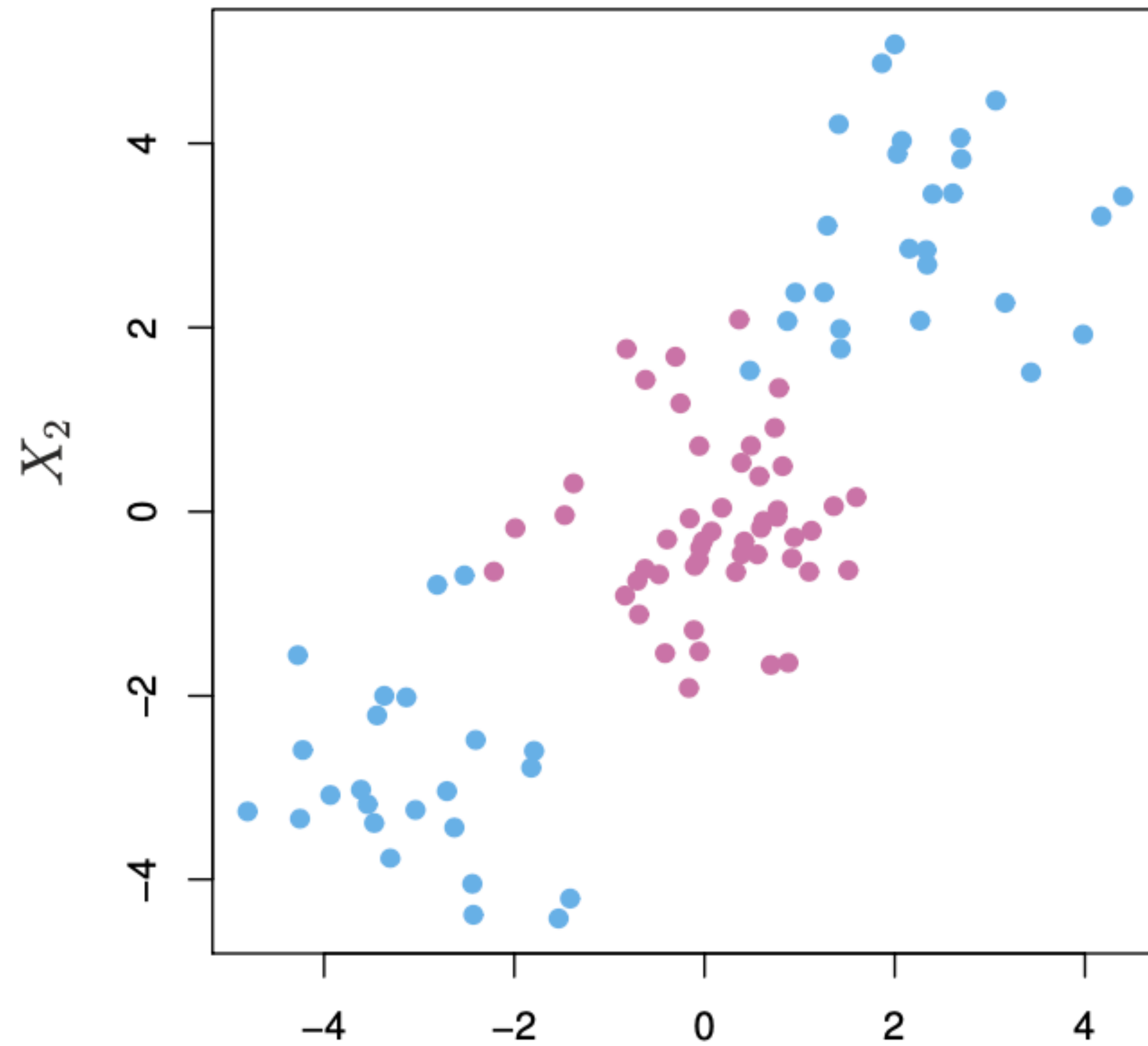


example



support vector machine

- what observation can you make in the following diagram?



- no linear boundary can be found
- what to do then?

- expand the feature space to get a quadratic, cubic or higher-order polynomial predictor
- e.g., fit the support vector in $2p$ features instead of p features

important!!!

- solution to the optimisation problem (slide 19) involves the inner product of the observations
- hint: can use the Lagrange multiplier to prove it

$$\langle X_l, X_{l'}' \rangle = \sum_{j=1}^p X_{lj} \cdot X_{l'j}$$

$$f(x) = \beta_0 + \sum_{l \in S}^n \alpha_l \langle x, x_l \rangle$$

Note that $\alpha_l = 0$ for any observation not a support vector

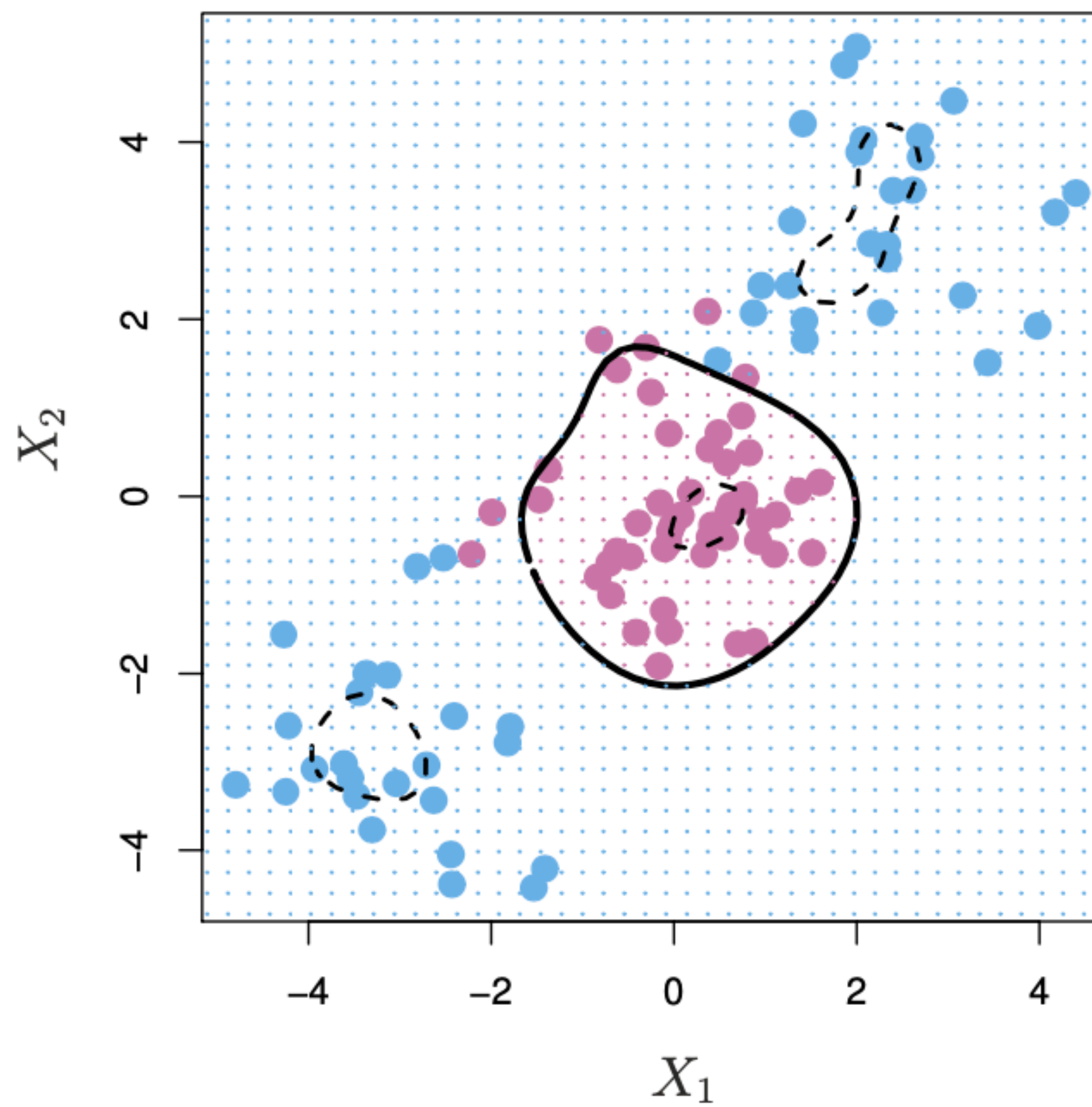
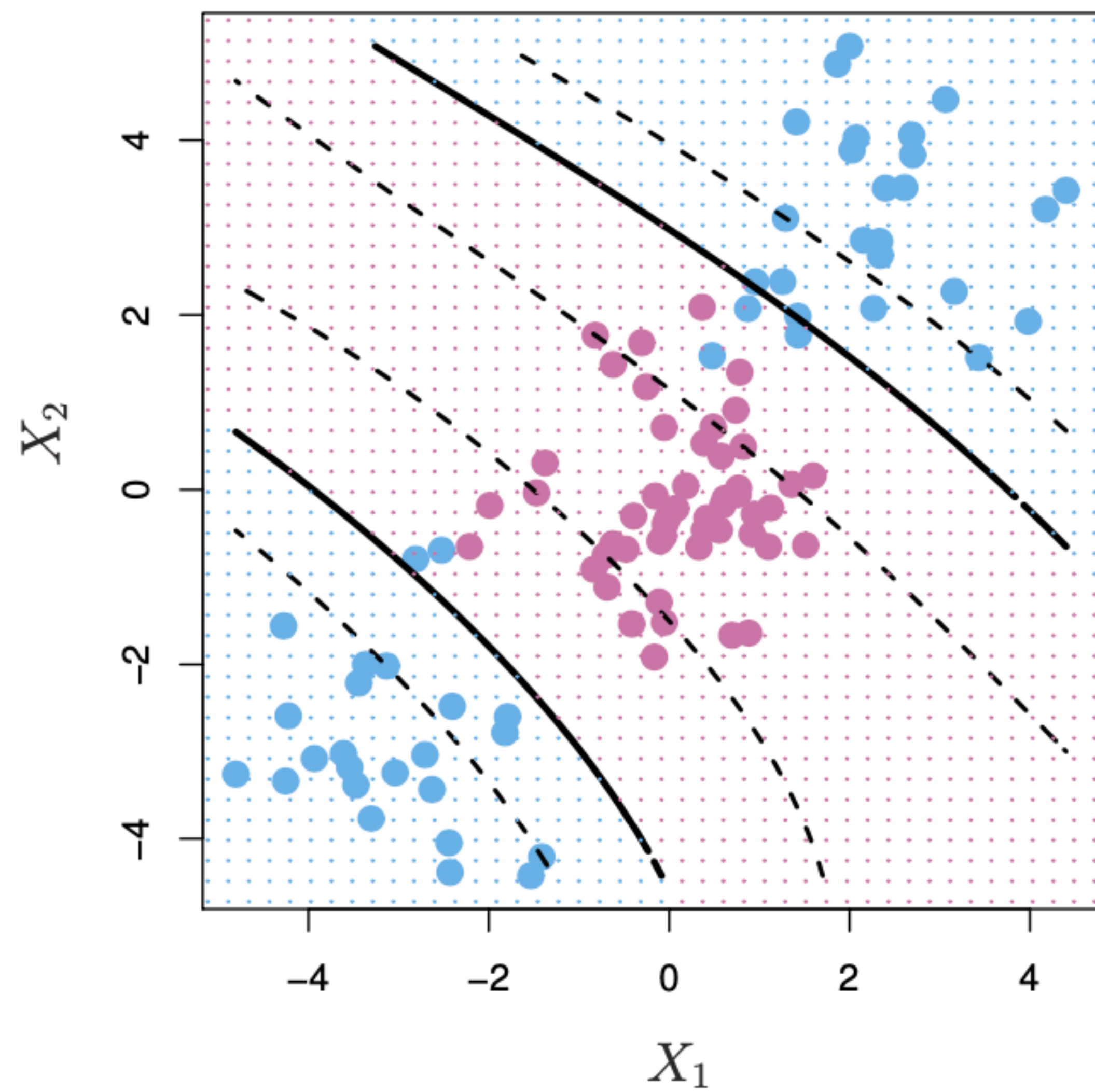
- can generalise the inner product with a kernel function
- kernel function quantifies the similarity between two observations
 - e.g., Pearson correlation for a linear kernel

kernel fns

$$K(x_l, x_{l'}) = (1 + \sum_{j=1}^p x_{lj} x_{l'j})^d$$

$$K(x_l, x_{l'}) = \exp(-\gamma \sum_{j=1}^p (x_{lj} - x_{l'j})^2)$$

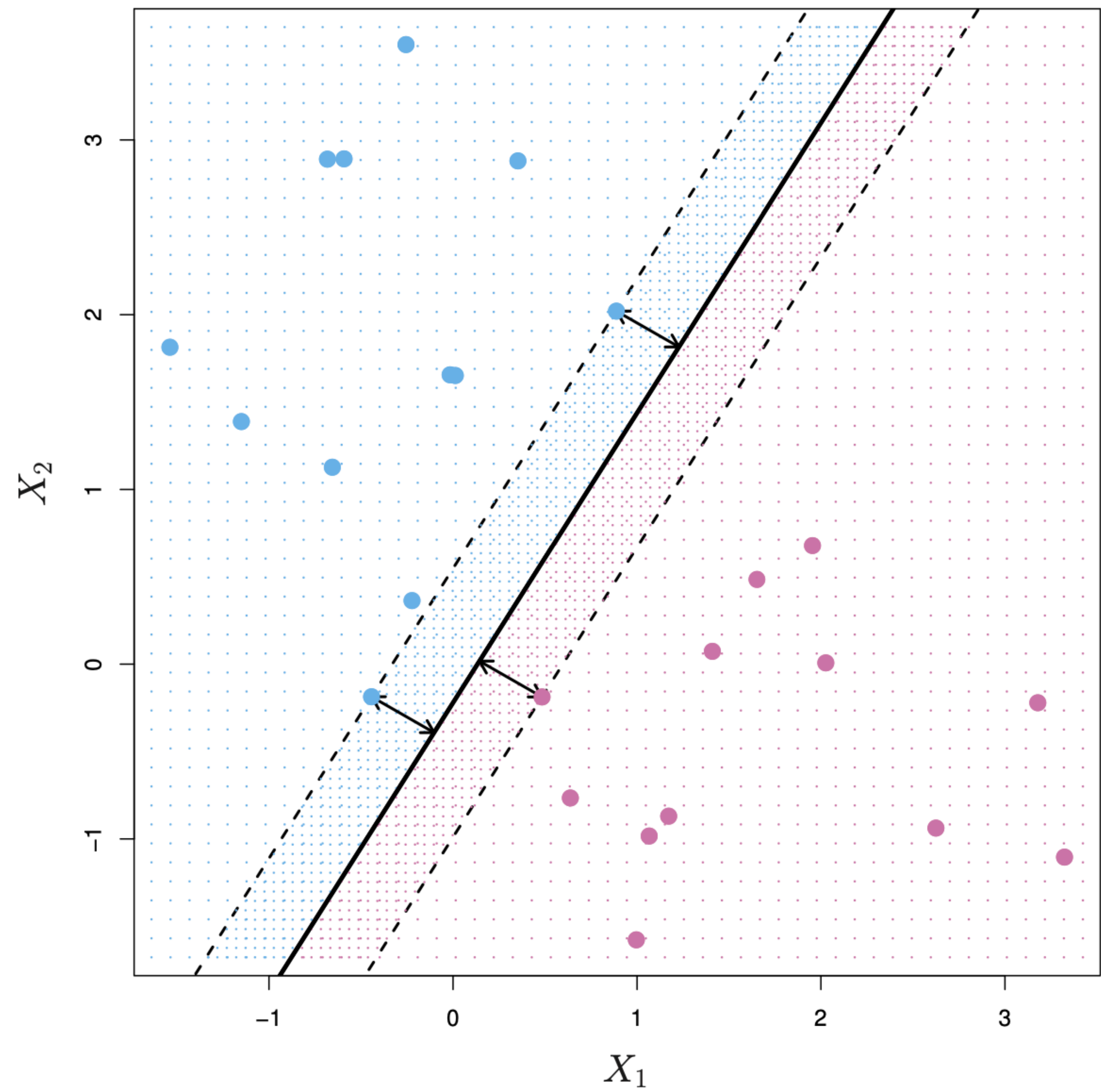
$$f(x) = \beta_0 + \sum_{l \in S} \alpha_l K(x, x_l)$$



> 2 classes?

- one vs one
 - construct svms comparing two classes and choose the most frequent
- one vs all
 - fit all svms and assign the observation to the class where the linear representation is the highest

regression



minimize

$$\beta_0, \beta_1, \dots, \beta_p$$

subject to

$$M$$

$$\sum_{j=1}^p \beta_j^2 = 1$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}) \geq M(1 - \epsilon_i - \xi_i)$$

$$\epsilon_i \geq 0; \xi_i \geq 0; \sum_{i=1}^N \epsilon_i \leq C$$

in short...

- maximal margin classifier
- support vector classifier
- support vector machine
- support vector regressor